

1 Introduction

Social media has become a major source of information for many Americans. Leading up to the 2016 US presidential election, around 14% of Americans indicated social media as their primary source of news (Allcott and Gentzkow (2017)), and by 2019, over 70% of Americans reported receiving at least *some* of their news from social media (Levy (2021)). At the same time, there is growing concern about viral misinformation in social media, including made-up news stories such as those claiming that there were no mass shootings under Donald Trump or that Hillary Clinton approved ISIS weapon sales.¹ Some recent evidence also suggests that misinformation on social media has impacted critical decisions such as vaccinations against COVID-19 (see Pennycook et al. (2018); Pennycook et al. (2020b)).

Although there is yet no consensus on what promotes the spread of falsehoods and misleading content on social media, two sets of factors have been emphasized. The first is the presence of echo chambers, which arise when individuals communicate and share content with like-minded users (Sunstein (2018), Lazer et al. (2018)). Törnberg (2018) and Vicario et al. (2016) show that echo chambers reinforce existing political viewpoints and tend to propagate misinformation. Social media, much more than traditional media, allows individual users to choose who and what they listen to, and thus echo chambers may be an unavoidable side effect. However, there is also evidence that echo chambers are a result of the “filter bubbles” that platform algorithms create (see Levy (2021) on Facebook). The second factor conjectured to have fueled misinformation is the general political polarization in many countries, and especially the United States,² and there is some preliminary evidence suggesting that polarization has contributed to selective exposure to questionable content on social media as well (Guess et al. (2018)). Despite the importance and salience of these issues, we do not currently have a framework to understand how online interactions impact the spread of misinformation and what factors shape incentives for sharing low-reliability content.

In this paper, we develop a parsimonious model of online sharing behavior in the presence of misinformation, and as a first step, we focus on the behavior of fully Bayesian agents.³ Our model is inhabited by a set of N agents, each of which has a prior about the state of the world (“ideological bias”), and is connected to the rest of the users via a network, which is given by agents’ friends and acquaintances, and is also shaped by the algorithms of the social media platform. A news article, defined by an underlying type (truthful or containing misinformation), a message (right-wing or left-wing), and a level of reliability (which determines the likelihood of misinformation), is then seeded at one of the agents. The message and the level of reliability of the article are common knowledge, while whether it is truthful or contains misinformation is unobserved, and agents form beliefs about this

¹See <https://www.snopes.com/fact-check/mass-shootings-under-trump/> and <https://www.cnn.com/2016/12/30/read-all-about-it-the-biggest-fake-news-stories-of-2016.html>, respectively.

²While there has been some debate about whether polarization has been mainly among politicians (see Fiorina et al. (2008) and Prior (2013)), there is considerable evidence that polarization has also risen among the general public (see Pew Research Center (2014) and Abramowitz (2010)).

³Myopic reactions and biased behavior appear to play some role, for example, via the “confirmation bias” in social media behavior (see, e.g., Buchanan (2020) and Pennycook and Rand (2019b)), but we believe that the Bayesian benchmark we construct already generates a number of empirically-relevant and rich results. We view incorporating realistic and relevant behavioral biases as a next step in this research agenda.